

What Determines Reported Savings in Carbon-Aware Virtual Machine Scheduling?

Capacity, Home Region, and Region-Set Composition

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Abstract

Reported reductions from carbon-aware virtual machine scheduling vary widely across the literature, from a few per cent to more than eighty. We show that much of this variation is produced not by the schedulers but by three choices in the evaluation that are seldom reported. Using CloudSim Plus with published hourly emission factors for seven grid regions and a production virtual machine trace, and adopting the host platform, simulator and baseline of a recent spatio-temporal scheduling study, we isolate each in turn. First, the value of deadline flexibility is conditional on load: widening the deadline margin from 6 h to 48 h reduces emissions by 32.1 ± 6.1 % at 13 % capacity utilisation but by 0.9 ± 0.6 % at 82 %. Second, the performance of a time-shifting policy is determined by the region in which its workload originates rather than by the temporal structure it exploits: the correlation with the origin region’s mean carbon intensity is between +0.941 and +0.998, while the correlation with its diurnal amplitude is indistinguishable from zero, and results across seven candidate origins span a factor of 3.6. Third, holding workload, scheduler and hardware fixed, one region set reports approximately twice the reduction of another at matched utilisation, because it contains a region 4.7 times cleaner than its nearest rival. A single mechanism underlies all three: emissions track the concentration of work into the cleanest reachable grid, with a correlation below -0.93 throughout, and the largest reductions we observe coincide with placing 95 % or more of the workload in one region. The results hold across five workload samples, two seasons, two workload classes and two host power models. We conclude with the four quantities an evaluation of carbon-aware scheduling should report if its figures are to be comparable.

Keywords: carbon-aware scheduling; cloud computing; virtual machine placement; deadline-constrained scheduling; evaluation methodology; CloudSim Plus

1 Introduction

The carbon intensity of electricity varies by hour and by location. Scheduling computation to exploit that variation, by deferring work to cleaner hours or by relocating it to cleaner grids, is now a well-established line of research, and the reductions reported for it are large. They are also inconsistent. Studies of temporal shifting report reductions that vary substantially with the region examined [19], and studies of combined spatial and temporal shifting report ranges wider still. Zanolto et al. [20], whose formulation this paper takes as its starting point, report a 79.25 % reduction against a carbon-agnostic baseline in their most permissive configuration and 13.46 % in their most restrictive one. Work examining the limits of these techniques has established that headline figures depend heavily on the assumptions behind them [17], but the parameters responsible have not been isolated individually.

The two configurations that produce the 13 % and the 79 % differ in two respects at once: the permissive one allows fifty simultaneous jobs per region where the restrictive one allows five,

and it draws on a different set of candidate regions. Neither parameter is varied alone, so the six-fold difference cannot be attributed to either. The same study is explicit that the per-region concurrency limit is consequential and that its estimation is left for future work. This paper takes up that question, and finds that the answer accounts for much of the range.

This paper argues that a substantial part of the variation is not a property of the schedulers at all. It is produced by three choices in the evaluation which are seldom reported, and each of which moves the reported figure by more than the difference between the policies being compared.

The first is capacity. A scheduler that defers a request to a cleaner hour can do so only if capacity is free in that hour. Evaluations that place a workload into an effectively unconstrained pool measure the benefit of flexibility under the most favourable condition available, and this condition is rarely stated. We find that widening the deadline margin from 6 h to 48 h reduces emissions by 32.1 % when the system runs at 13 % utilisation and by 0.9 % when it runs at 82 %, a difference of more than thirty-fold produced entirely by how heavily the system is loaded.

The second is the origin region assigned to a time-shifting policy. Such a policy holds work in one region and moves it only in time, so its result is partly a statement about that region. We find that the correlation between the mean carbon intensity of the origin region and the emissions the policy achieves is between +0.941 and +0.998, whereas the correlation with the region’s diurnal amplitude, the quantity the policy actually acts upon, is indistinguishable from zero. Across the seven candidate origins in our primary region set, the same policy yields results spanning a factor of 3.6. A single reported figure is one draw from that spread, and the conventional choice of origin selects the cleanest available grid.

The third is the composition of the candidate region set. Running an identical workload and an identical scheduler at matched utilisation over two region sets, we find that one reports approximately twice the reduction of the other at every operating point. The set that reports more contains a region 4.7 times cleaner than its nearest rival, so that a single destination is optimal for nearly every request; the set that reports less does not.

These are not competing explanations for the spread in the literature but complementary ones, and they share a mechanism. Across every configuration we examine, the correlation between the fraction of the workload placed in the busiest region and total emissions is below -0.93 . Carbon-aware placement policies work by concentrating load into the cleanest reachable grid; capacity governs how much can be concentrated, the origin region governs where a time-shifting policy starts from, and the region set governs how much is to be gained by concentrating at all. The reductions these policies report are, to first approximation, measurements of the concentration they achieve.

The consequence for practice is that the largest reductions are obtained in precisely the configurations least likely to be deployable. At the lowest utilisation we examine, the policies achieving the largest reductions place between 95 % and 98 % of the workload in a single region.

Our contributions are as follows.

- We isolate the interaction between capacity utilisation and deadline margin, and show that the value of deadline flexibility is conditional on load rather than being a property of the workload or the scheduler (Section 5.1).
- We show that the reported performance of a time-shifting policy is determined by its origin region rather than by the temporal structure it exploits, and quantify the resulting spread (Section 5.2).
- We show that reported savings scale with the degree to which the candidate region set contains a dominant clean region, holding workload, scheduler and hardware fixed (Section 5.3).

- We establish that these results hold across five workload samples, two seasons, two workload classes, two region sets and two host power models, and we report the accounting basis on which each is measured (Section 5.4).

We adopt the host platform, the simulator, the carbon-agnostic baseline, the per-region concurrency constraint and the deadline-margin levels of Zanotto et al. [20], so that differences between their reported figures and ours cannot be attributed to any of those. Where we depart from them is in varying one parameter at a time, in sweeping the origin region of the time-shifting policy rather than fixing it, and in reporting emissions on an explicitly stated accounting basis. The evaluation uses published hourly emission factors [7] and a production virtual machine trace [2], and each experimental cell is checked to confirm that every virtual machine executed in the region its planner selected.

2 Related work

2.1 Temporal shifting

Deferring flexible work to hours of lower carbon intensity was examined systematically by Wiesner et al. [19], who characterised delay-tolerant workloads and quantified the potential for temporal shifting in four regions over a year. Their finding that the achievable reduction depends strongly on the region examined is the starting point for Section 5.2: we ask which property of the region is responsible, and find that it is the region’s mean intensity rather than its temporal structure. Earlier work on scheduling around on-site renewable supply [4, 5] addresses a related problem, though with generation local to the datacentre rather than grid intensity as the varying quantity.

2.2 Spatial and spatio-temporal shifting

Relocating work between grids has been studied both as load balancing across geo-distributed datacentres [21] and, more recently, as carbon-aware placement in production settings [11, 14]. Sukprasert et al. [16, 17] examine the limits of both temporal and spatial shifting and conclude that reported benefits are sensitive to the assumptions under which they are measured. Our work is complementary: where they establish that the assumptions matter in aggregate, we isolate three specific parameters and quantify each separately, holding the remainder fixed.

Zanotto et al. [20] evaluate combined spatial and temporal shifting for non-preemptible virtual machines under a Kubernetes-based architecture. They formulate placement as a binary integer program solved one request at a time against a greedily updated occupancy matrix, and evaluate it in CloudSim Plus against a carbon-agnostic round-robin baseline, with a per-machine power profile drawn from SPECpower and virtual machine requests sampled from a production trace. Their formulation is the direct antecedent of the policies compared in Section 5, and we adopt their host platform, simulator, baseline, per-region concurrency constraint and deadline-margin levels so that our results and theirs differ in as few respects as possible.

Their constraint limiting each region to M_j simultaneous jobs is introduced explicitly to prevent what they term the “black hole” case, in which the optimiser assigns the entire workload to the single cleanest region. They observe that the limit governs the outcome, that a permissive limit concentrates nearly all work in one region while a restrictive one distributes it, and that the value of M_j must be determined experimentally. They leave that estimation as future work. Sections 5.1 and 5.3 address it, and find that the parameter accounts for a larger share of the variation in reported savings than the scheduling policy does.

2.3 Carbon intensity data and its representation

The choice of carbon intensity signal is itself consequential. Maji et al. [9] show that location-based and market-based estimation methods can lead to different conclusions about the efficacy of the same optimisation. Forecast-based scheduling depends further on prediction quality, for which day-ahead [8] and multi-day [7] methods have been developed. We use published lifecycle emission factors throughout and plan against ground truth, so that the parameters under study are not confounded with prediction error; the effect of forecast quality is a separate question that this paper deliberately does not address. We note that Zanotto et al. [20] report closely comparable results under forecast and under perfect foresight, which supports treating prediction error as separable from the parameters studied here.

Infrastructure for carbon-aware experimentation has also received attention. Testbeds such as Vessim [18] and virtualised energy systems such as Ecovisor [13] provide environments in which carbon-aware applications can be developed and evaluated. Our aim differs: we do not propose a new environment but examine what an evaluation conducted in such an environment is measuring.

2.4 Simulation and power modelling

CloudSim [1] and its successor CloudSim Plus [12] are widely used for cloud resource-management studies, and we adopt the latter. Reported energy figures in such studies depend on the host power model, and Ismail and Materwala [6] survey the models in use and their accuracy. We use a measured SPECpower_{ssj2008} curve for the specific host modelled [15] and, because the choice is consequential, repeat the entire evaluation under a linear approximation to establish which of our conclusions depend on it (Section 5.4).

2.5 Position of this work

Prior work has proposed carbon-aware scheduling policies and has examined their limits. What has not been established is how much of the variation in reported savings is attributable to the evaluation configuration rather than to the policies themselves. This paper addresses that question directly, and the answer bears on how such evaluations should be reported.

3 Method

3.1 Overview

We evaluate carbon-aware placement of non-preemptible virtual machines across geographically distributed datacentres. A workload of independent VM requests arrives over a 720 h horizon. Each request must be assigned to one region and one start hour, subject to a per-region concurrency cap and a per-request deadline. Once started, a VM runs to completion in the region where it was placed; migration and preemption are outside the scope of this study, following the assumptions of Zanotto et al. [20].

The evaluation is implemented on CloudSim Plus 8.5.7 [12] under Java 17. Placement decisions are taken by a planner operating on an occupancy ledger, and the resulting schedule is then executed in the simulator so that host-level power draw, and hence emissions, are obtained from simulated execution rather than from the planner’s own cost model. This separation matters: the planner minimises a carbon-intensity objective, whereas the reported emissions depend additionally on how many physical hosts the schedule activates. Sections 3.4 and 3.6 describe the accounting and the checks that connect the two. Figure 1 summarises the pipeline.

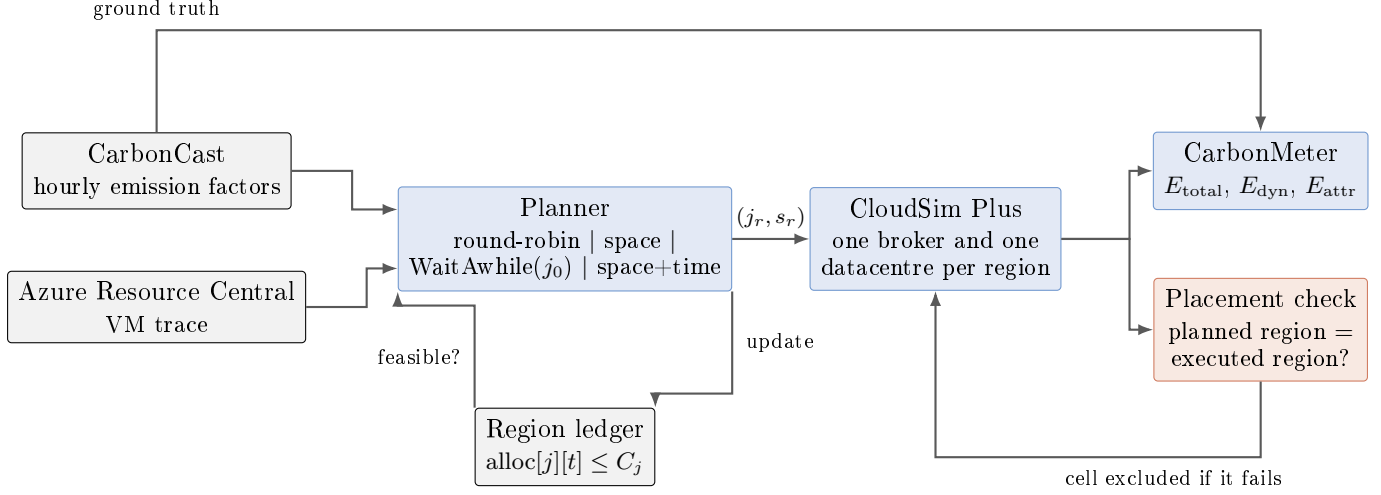


Figure 1: Evaluation pipeline. The planner assigns each request a region and a start hour subject to the per-region concurrency cap, and the resulting schedule is then executed in the simulator, so that emissions are obtained from simulated host power rather than from the planner’s own objective. Every cell is checked after execution to confirm that each virtual machine ran in the region its planner selected; cells that fail are excluded rather than reported. The carbon trace enters twice, once as the planning signal and once as the measurement signal, which are identical here and separable when forecast quality is under study.

3.2 System model

Let J denote the set of eligible regions and $T = \{0, \dots, H - 1\}$ the hours of the horizon, with $H = 720$. Each request r is a tuple $(p_r, m_r, d_r, a_r, \delta_r)$ giving its core count, memory, duration in hours, arrival hour and deadline hour. A placement assigns r to a region $j_r \in J$ and a start hour $s_r \geq a_r$. The request meets its deadline when $s_r + d_r \leq \delta_r$.

Regional capacity is expressed as an occupancy matrix $\text{alloc}[j][t]$ counting VMs running in region j at hour t , subject to

$$\text{alloc}[j][t] \leq C_j \quad \forall j \in J, \forall t \in T, \quad (1)$$

where C_j is the per-region concurrency cap. Constraint (1) represents the finite capacity a tenant can command in any single region and is the parameter whose effect this paper isolates.

Each region hosts one datacentre containing an identical fleet of homogeneous hosts. The host is a Dell PowerEdge XR8620T with an Intel Xeon Gold 6433N (32 cores at 2.00 GHz) and 256 GB of memory. This is the platform modelled by Zanotto et al. [20], who likewise draw its power profile from SPECpower and likewise evaluate on CloudSim Plus against a carbon-agnostic round-robin baseline; adopting the same host removes hardware as a source of difference between their reported figures and ours. Each core is modelled as a processing element of 2000 MIPS. A VM requests p_r processing elements and m_r memory, and executes a single cloudlet of length $2000 \cdot d_r \cdot 3600$ MI at full CPU utilisation, so that its runtime equals d_r hours.

3.3 Host power model

Host power is taken from the published SPECpower_ssj2008 result for the same machine [15] (Dell Inc., PowerEdge XR8620T with Xeon Gold 6433N, tested 15 July 2024, 6564 overall ssj_ops/watt). The sensitivity of simulated energy figures to the power model adopted is well documented [6]. Because the simulated host matches the benchmarked one in core count, clock and memory, the measured curve is used without rescaling. Table 1 reports it.

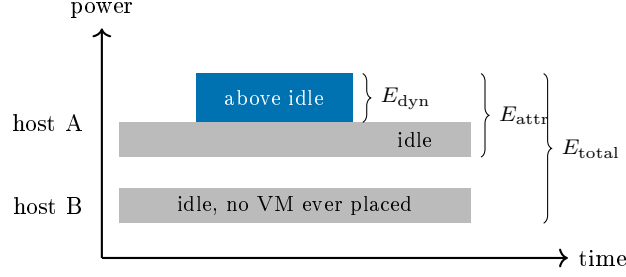


Figure 2: The three accounting bases, shown schematically for a two-host fleet. E_{total} charges every provisioned host for the whole window and is therefore invariant to placement in a fixed fleet. E_{dyn} charges only draw above idle and is unaffected by how many hosts a schedule activates. E_{attr} charges the full draw of hosts that carry load and nothing for hosts that do not, so it responds both to regional placement and to consolidation. Host B is provisioned in all three cases; it contributes to E_{total} alone.

Table 1: Measured host power at decile load points (SPECpower_{ssj2008}).

Load (%)	0	10	20	30	40	50	60	70	80	90	100
Power (W)	230	244	260	273	301	340	372	385	400	416	420

Power at intermediate utilisations is obtained by linear interpolation between adjacent measured points, since SPEC publishes no finer resolution. Two properties of this curve affect the results. First, the machine draws 230 W at idle against a 420 W peak, an idle fraction of 55 %; the range a scheduler can influence is therefore 190 W per host. Second, the curve is not linear: it is nearly flat between 0 % and 30 % load, rises steeply between 30 % and 60 %, and flattens again above 70 %. A linear model fitted only to the idle and peak points misprices both ends of the range traversed by the capacity sweep. Section 4.4 describes how sensitivity to this choice is tested.

3.4 Carbon accounting

Emissions are obtained by integrating host power against regional carbon intensity. Sampling occurs on every simulation clock tick, with a datacentre scheduling interval of 300 s, and each interval is charged at the carbon intensity of the hour in which it begins. For a host h in region j with utilisation $u_h(t)$, the instantaneous rate is $P(u_h(t)) \cdot \text{CI}_j(t)$, where P is the interpolated curve of Table 1 and $\text{CI}_j(t)$ the regional carbon intensity in gCO₂eq/kWh.

We report emissions on three bases, illustrated in Figure 2, because the choice determines what a comparison can detect.

$$E_{\text{total}} = \sum_{j \in J} \sum_{h \in \mathcal{H}_j} \int P(u_h(t)) \text{CI}_j(t) dt, \quad (2)$$

$$E_{\text{dyn}} = \sum_{j \in J} \sum_{h \in \mathcal{H}_j} \int [P(u_h(t)) - P(0)] \text{CI}_j(t) dt, \quad (3)$$

$$E_{\text{attr}} = \sum_{j \in J} \sum_{h \in \mathcal{H}_j} \int \mathbb{1}\{u_h(t) > 0\} P(u_h(t)) \text{CI}_j(t) dt. \quad (4)$$

Equation (2) is the provider’s meter: every provisioned host draws power for the whole measurement window whether or not it carries load. Because the same fleet exists in the same

regions under every policy, the idle component of E_{total} is invariant to placement. In our configuration this component dominates, and two policies with entirely different schedules report totals differing by less than 2 %. A study reporting only E_{total} cannot distinguish the policies it compares.

Equation (3) counts only power above idle. It is sensitive to placement and is unaffected by how many hosts a schedule activates, which makes it the appropriate basis when the quantity of interest is the carbon cost of the work itself.

Equation (4) charges the full power of hosts that carry at least one VM and nothing for hosts that carry none. This is the consumer-side basis: a tenant is accountable for the machines its workload occupies, not for the provider’s idle capacity. It is sensitive both to regional placement and to consolidation, and it is the basis we report as primary, with E_{dyn} given alongside so that the consolidation component of any reduction remains visible.

All three are normalised by admitted VM-hours to permit comparison across configurations.

3.5 Scheduling policies

Four policies are compared. All see the same carbon trace and the same occupancy ledger, so differences arise from the decision rule alone.

Round robin places each request at its arrival hour in the next region with available capacity, ignoring carbon intensity. It is the carbon-agnostic baseline and is included so that the share of any reported reduction attributable to baseline weakness remains visible.

Space places each request at its arrival hour in the eligible region with the lowest mean carbon intensity over the request’s execution window. It shifts across regions but never defers.

WaitAwhile, after Wiesner et al. [19], holds each request in its origin region and defers it to the cleanest feasible start hour within its deadline. The origin region is a parameter of the policy, denoted $\text{WaitAwhile}(j_0)$; Section 4.3 explains why it is swept rather than fixed.

Space+time searches jointly over regions and start hours within the deadline. Zanolto et al. [20] solve the corresponding problem as a binary integer program applied to one request at a time against a greedily updated occupancy matrix. For a single request the feasible set has $|J| \times |T|$ members, so exhaustive enumeration returns the same optimum without a solver dependency.

Every policy shares a common fallback: when no feasible slot exists within the deadline, the request is placed at the earliest feasible slot anywhere, accepting a deadline violation. The fallback searches all regions, including for *WaitAwhile*, since a policy that declines to relocate once its deadline is already unreachable would be constrained beyond what its formulation states. Requests that cannot be placed even beyond their deadline cause the run to fail rather than to be silently dropped.

3.6 Placement validity

Two properties of CloudSim Plus can silently break the correspondence between the planned schedule and the executed one. A datacentre that cannot accommodate a VM causes the broker to reroute it to another datacentre, so that the executed placement differs from the planned region; and the broker’s datacentre mapper is applied per batch of waiting VMs rather than per VM, so that a single broker cannot express per-VM regional assignment at all. We therefore instantiate one broker per region, each pinned to its own datacentre, which makes regional misplacement impossible by construction.

After each run we verify, for every VM, that it was created in the datacentre its planner selected and that its cloudlet completed within the measurement window. Placement is recovered from each broker’s list of created VMs rather than from the VM’s own creation flag, which reads false once a VM has been destroyed at the end of its execution. Cells failing this check are

Table 2: Regional characteristics over the 720 h window from 1 January 2020. Mean carbon intensity in gCO₂eq/kWh; diurnal amplitude as a percentage of the regional mean.

Set	Region	Mean CI	Diurnal (%)
US	BPAT	63	14.3
	NYISO	209	25.6
	ISNE	297	14.5
	CISO	299	52.7
	PJM	341	7.8
	ERCO	354	24.0
	FPL	359	21.1
EU	SE	39	1.7
	ES	183	13.7
	DE	323	15.3
	NL	483	8.4
	PL	663	6.1

excluded from analysis rather than reported, since their emissions do not describe the schedule under test.

4 Experimental design

4.1 Carbon intensity traces

Carbon intensity is taken from CarbonCast [7], which publishes hourly lifecycle emission factors for thirteen grid regions across North America, Europe and Australia for 2020–2021. We use it in preference to commercial sources because the data may be redistributed, which allows the experiment to be reproduced from a public archive, unlike the metered access of commercial providers [3], and because it also publishes paired actual and forecast series, which the companion study requires.

All thirteen regional files were verified before use: each contains 17544 hourly observations with no missing hours, no duplicate timestamps and no non-positive values. Timestamps are not written in a uniform format across regions, and files are aligned on a common UTC window rather than on each file’s own first row, since indexing each region independently would offset the regions against one another.

Zanotto et al. [20] site their clusters at Azure cloud regions. We instead use grid regions for which independently published hourly emission factors exist, since the object of study is the effect of region-set composition and that requires regional carbon intensity to be a measured input rather than a modelling choice. Two region sets are used. The primary set comprises seven United States independent system operators; the second is a five-region European set retained as a control. Table 2 reports their characteristics over the 720 h window beginning 1 January 2020. The column headed *diurnal* gives the spread of hour-of-day means as a percentage of the regional mean, which isolates the repeatable daily cycle available to a deferral policy from one-off excursions.

The two sets differ in a way that is central to Section 4.3. In the US set the cleanest region is 3.3 times cleaner than the next cleanest, and the full spread is 5.7-fold. In the EU set the cleanest region is 4.7 times cleaner than the next, and the spread is 17.1-fold. The EU set therefore admits a dominant optimum, and is retained precisely so that the consequences of that dominance can be measured.

4.2 Workload

VM requests are sampled from the Azure Resource Central trace [2, 10], which records 2 695 549 virtual machines with creation and deletion timestamps, core count, memory and a workload category assigned by the provider. Timestamps run to 2 591 400 s, exactly the 720 h horizon simulated here, so no rescaling of arrival times is required.

Three filters are applied. Virtual machines still running at the end of the trace (224 249 records) are right-censored and are excluded, because treating the truncation point as a completion would systematically shorten the long-running VMs that deferral is intended to target. Virtual machines shorter than two hours are excluded, since a VM shorter than the scheduling granularity cannot meaningfully be deferred. Sampling is by reservoir rather than by taking a prefix of the file, since the trace is not shuffled.

A sample of 1 000 requests drawn under these filters yields approximately 33 000 VM-hours over the horizon, a mean concurrency of about 46 VMs and a mean VM lifetime of 34 h. The provider’s categories are unevenly distributed: 2 446 819 records are labelled *Unknown*, 159 497 *Delay-insensitive* and 78 309 *Interactive*. A separate sample restricted to the delay-insensitive class is used as a robustness condition. That class has a mean lifetime of 340.7 h, an order of magnitude longer than the unrestricted mixture, so the sample size is reduced to 100 requests to hold total demand, and hence capacity utilisation, constant across the two conditions.

4.3 Factors and levels

Four factors are varied.

Capacity. Total concurrent slots take four levels, distributed evenly across the regions of the set. For the US set these are 56, 84, 140 and 350 slots, corresponding to capacity utilisations of 82 %, 55 %, 33 % and 13 % respectively; the EU levels are chosen to match those utilisations. The sweep is defined on total slots rather than on a per-region cap because a per-region cap makes total capacity depend on the size of the region set, so that a policy confined to one region would compete against policies commanding $|J|$ times more hardware.

Deadline margin. The slack allowed beyond a request’s own duration takes values of 6, 12, 24 and 48 h.

Home region. WaitAwhile is instantiated once per region of the set rather than once per configuration. A single instantiation reports the performance of one arbitrary origin, and in both region sets the conventional choice of the first listed region selects the cleanest grid available, which assigns the policy the optimum without stating that it has done so.

Region set. The US and EU sets described above.

Each configuration is repeated across five independent workload samples for the US set and three for the EU set, obtained by varying the reservoir sampling seed with all other inputs held fixed. Reported spreads are standard deviations across these repetitions and represent workload sampling variability.

4.4 Controls

Several quantities are deliberately held constant so that the factors above are not confounded.

The physical fleet is sized once from the largest capacity level in the sweep and held fixed across all cells. Sizing each cell to its own cap would make total emissions a function of the capacity axis through the quantity of hardware provisioned rather than through scheduling. Fleet size is derived from the largest VM in the workload rather than the mean, because the cap bounds the number of concurrent VMs and not their size, and CloudSim Plus allocates from individual hosts rather than from a pooled resource.

All policies are measured over an identical window of $H + 48$ hours. Without a fixed window a policy that defers work would run for longer and accrue additional idle energy, which would

Table 3: Attributed emissions (kgCO₂eq) for space+time scheduling, mean across five workload samples with standard deviation in parentheses. Utilisation is total demand divided by total available slot-hours.

Capacity	Utilisation	Deadline margin (h)			
		6	12	24	48
56	82 %	2065 (138)	2067 (131)	2059 (131)	2047 (134)
84	55 %	1821 (117)	1787 (128)	1755 (128)	1718 (127)
140	33 %	1421 (102)	1390 (107)	1350 (108)	1321 (110)
350	13 %	683 (88)	555 (119)	484 (95)	463 (71)

be read as higher emissions irrespective of its placement decisions. The ledger extends a further 336 h beyond the horizon so that deadline-violating fallbacks always have a feasible slot.

Planning is performed against the ground-truth carbon trace throughout, so the policies operate under perfect foresight. This is intentional. Forecast error would interact with both capacity and deadline margin, and the object of this study is the effect of those two parameters in isolation; the effect of prediction quality is addressed separately. Carbon-intensity forecasts of the kind required for that analysis are themselves an active subject [7, 8].

Finally, the entire experiment is repeated under a linear power model with a 120 W idle and 480 W peak, an idle fraction of 25 % against the 55 % of the measured curve, in order to establish whether the reported effects depend on the power model adopted.

5 Results

All 320 cells of the US sweep passed the placement check of Section 3.6: every virtual machine executed in the region its planner selected, and every cloudlet completed within the measurement window. Figures are reported on the attributed basis of Eq. (4) unless stated otherwise, with the dynamic basis of Eq. (3) given alongside where the two diverge. Spreads are standard deviations across the five workload samples.

5.1 Capacity utilisation determines the value of deadline flexibility

Table 3 reports emissions for the joint space-and-time policy across the capacity and deadline-margin grid. Reading down a column gives the effect of capacity at fixed flexibility; reading across a row gives the effect of flexibility at fixed capacity.

The two effects differ by more than an order of magnitude. Increasing capacity from 56 to 350 slots reduces emissions by $77.4\% \pm 2.3$ at the widest deadline margin, and the reduction is between 66.1 % and 77.4 % at every margin considered. Increasing the deadline margin from 6 h to 48 h reduces emissions by $32.1\% \pm 6.1$ at the lowest utilisation but by only $0.9\% \pm 0.6$ at the highest.

The value of deadline flexibility is therefore not a property of the workload or of the scheduler alone; it is conditional on how heavily the system is loaded. Table 4 states this dependence directly.

At 82 % utilisation the effect of deadline margin is 1.5 standard deviations from zero and is not distinguishable from no effect at any conventional threshold. The interpretation is mechanical rather than statistical: deferring a request is only useful if a cleaner slot is free when the request is ready to move, and under heavy load the cleaner slots are already occupied. Flexibility that cannot be exercised has no value.

The saturated row of Table 3 carries a second qualification. At 56 slots, between 152 and 408 of the 1 000 requests miss their deadlines depending on policy and margin, whereas at 84 slots

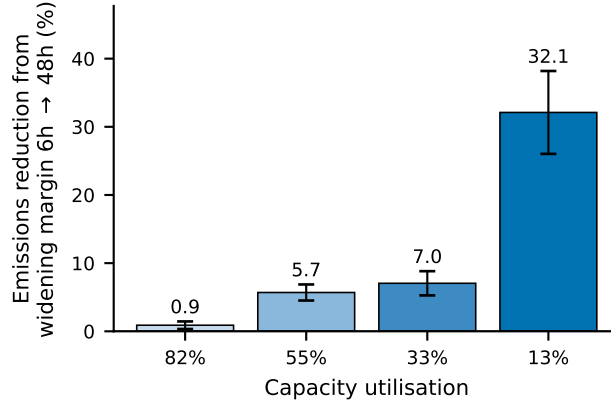


Figure 3: The value of deadline flexibility is conditional on load. Bars show the reduction in attributed emissions obtained by widening the deadline margin from 6 h to 48 h under space+time scheduling; error bars are one standard deviation across five workload samples.

Table 4: Reduction in attributed emissions obtained by widening the deadline margin from 6 h to 48 h, by capacity utilisation. Mean and standard deviation across five workload samples.

Capacity utilisation	Reduction from 6 h to 48 h margin
82 %	0.9 % $\pm$ 0.6
55 %	5.7 % $\pm$ 1.2
33 %	7.0 % $\pm$ 1.8
13 %	32.1 % $\pm$ 6.1

and above no request does. Emissions comparisons in the saturated regime therefore compare schedules that are all failing, and we report that row as a boundary condition rather than as an operating point.

5.2 Time shifting is dominated by the choice of origin region

WaitAwhile [19] defers requests within a fixed origin region. Its reported performance therefore depends on which region that is. We instantiate the policy once per region of the set and examine what predicts the outcome.

Across all four capacity and margin combinations, the correlation between the mean carbon intensity of the origin region and the emissions the policy achieves is between +0.941 and +0.998, with a standard deviation across workload samples of at most 0.017. The correlation between the origin region’s diurnal amplitude and the same emissions is between -0.066 and $+0.085$, in every case within one standard deviation of zero. Table 5 reports both.

The policy’s outcome is thus determined by how clean its origin grid is and not by how much daily variation that grid offers, even though daily variation is the only quantity the policy acts upon. The US set contains a region, CISO, whose diurnal amplitude of 52.7% is more than six times that of the flattest region in the set; instantiating WaitAwhile there yields no advantage over instantiating it in a region of comparable mean intensity and a quarter of the amplitude.

The practical consequence is the spread in Table 6. At the lowest utilisation, the best origin region yields 467 kgCO₂eq and the worst 1705, a factor of 3.6. Space shifting, which is free to choose its region per request, yields 701. The best available origin outperforms space shifting; the median origin is worse than it by a factor of two.

A single reported figure for a time-shifting policy is therefore a draw from a distribution whose width exceeds the difference between the policies being compared, and the draw is determined by a parameter that evaluations do not customarily state. Prior work on the limits of temporal

Table 5: What predicts WaitAwhile’s emissions. Pearson correlation between the policy’s attributed emissions and two properties of its origin region, across the seven regions of the US set. Mean and standard deviation over five workload samples.

Capacity	Margin (h)	Origin mean CI	Origin diurnal amplitude
84	6	$+0.953 \pm 0.017$	-0.066 ± 0.067
84	48	$+0.941 \pm 0.016$	-0.007 ± 0.076
350	6	$+0.989 \pm 0.006$	$+0.056 \pm 0.026$
350	48	$+0.998 \pm 0.002$	$+0.085 \pm 0.020$

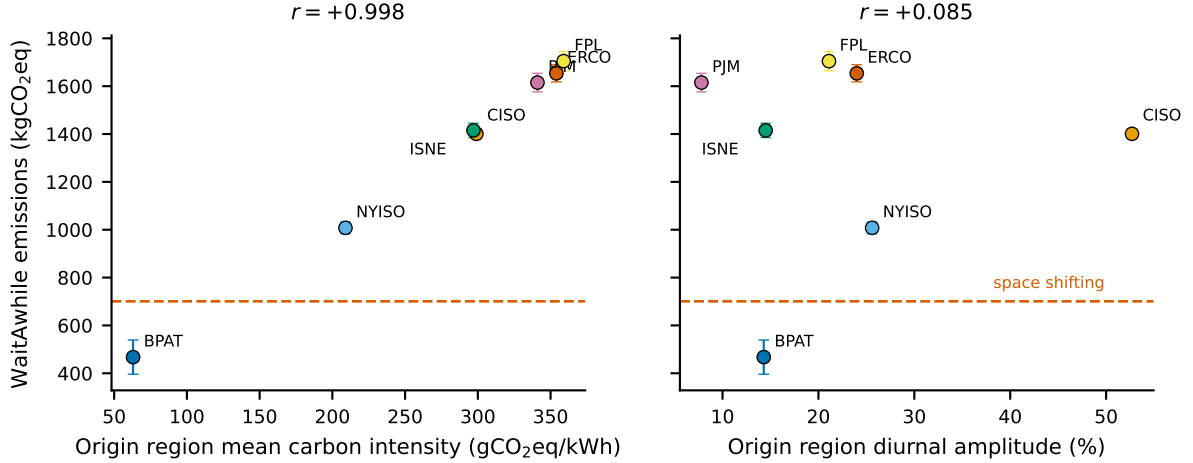


Figure 4: WaitAwhile emissions against two properties of the origin region, at 350 slots and a 48 h margin. Each point is one of the seven candidate origins, averaged over five workload samples; error bars are one standard deviation. The dashed line marks space shifting in the same cell. The policy’s outcome tracks the origin’s mean carbon intensity (left) and not its diurnal amplitude (right), although the latter is the only quantity the policy acts upon.

shifting [17] does not isolate this parameter. In both region sets examined here, the conventional choice of the first region in the candidate list happens to select the cleanest grid available, which assigns the policy the most favourable draw.

5.3 Reported savings scale with the composition of the region set

The same workload and the same scheduler were run over two region sets at matched capacity utilisation. Table 7 reports the reduction that space shifting achieves relative to the carbon-agnostic baseline in each.

The EU set reports roughly twice the reduction of the US set at every operating point, and the difference exceeds the combined standard deviations in all four rows. Nothing about the scheduler, the workload or the amount of hardware differs between the two columns. The difference is the composition of the candidate set: the EU set contains a region 4.7 times cleaner than its nearest rival, so that a single destination is optimal for almost every request, whereas in the US set the corresponding ratio is 3.3 and the placement decision remains contested.

Regional concentration confirms the mechanism, and Figure 6 shows it directly. Across all valid cells of both sets, the correlation between the share of the workload placed in the busiest region and attributed emissions is -0.930 for the US set and -0.963 for the EU set. In the EU set at the lowest utilisation, space shifting places 98 % of the workload in a single region; in the US set it places 96 %. The reductions these policies report are, to a first approximation, a measure of how much of the workload the capacity constraint permits them to move into the

Table 6: WaitAwhile attributed emissions (kgCO₂eq) across the seven possible origin regions, with space shifting for comparison. Values are means over five workload samples.

Capacity	Margin (h)	Best origin	Median origin	Worst origin	Space
84	6	1841	1974	2039	1857
84	48	1781	1905	1984	1857
350	6	683	1432	1698	701
350	48	467	1415	1705	701

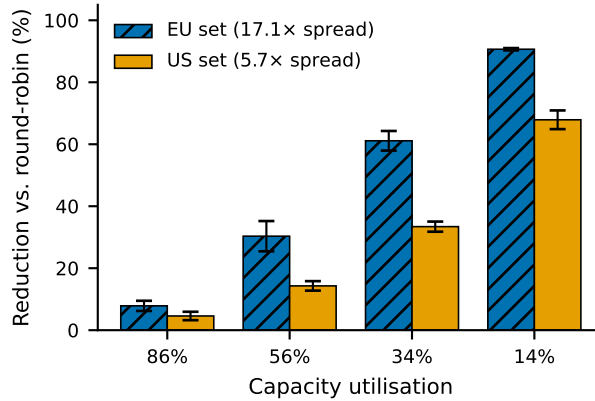


Figure 5: Reported savings scale with the composition of the candidate region set. Identical workload, scheduler and hardware; the sets differ only in the ratio between their cleanest region and the next cleanest. Error bars are one standard deviation across workload samples.

cleanest available grid.

5.4 Robustness

Season. Repeating the sweep on a 720 h window beginning in July rather than January changes the correlation between origin carbon intensity and WaitAwhile’s emissions from the range $[+0.941, +0.998]$ to $[+0.760, +0.892]$, and the correlation with diurnal amplitude from approximately zero to the range $[-0.518, -0.285]$. Diurnal structure therefore acquires a secondary influence in summer, when solar generation widens the daily cycle in several regions, without displacing origin cleanliness as the dominant term. The ordering of policies is unchanged.

Workload class. Restricting the workload to the class the provider labels delay-insensitive preserves the result, with the correlation between origin carbon intensity and emissions at $+0.822$, $+0.920$ and $+0.936$ in three of four cells. The exception is the combination of tight capacity and wide margin, where the correlation with origin intensity falls to $+0.403$ and that with diurnal amplitude rises to $+0.776$: when capacity constrains relocation and the deadline permits deferral, temporal structure is the only remaining lever and briefly becomes the dominant one.

This class also differs from the general mixture in a way that bears on the premise of deadline-based scheduling. Its mean virtual machine lifetime is 340.7 h against 34 h for the unrestricted sample, an order of magnitude longer. A 48 h margin therefore represents 141 % of the mean lifetime in the general mixture but only 14 % in the class explicitly designated as tolerant of delay. The population with the most permission to wait has the least relative room in which to do so.

Table 7: Reduction in attributed emissions from space shifting relative to round-robin placement, at matched capacity utilisation, for two region sets. Mean and standard deviation across workload samples (five for US, three for EU).

Capacity utilisation	EU set	US set
86 %	7.8 % $\pm$ 1.6	4.6 % $\pm$ 1.4
56 %	30.3 % $\pm$ 4.9	14.3 % $\pm$ 1.5
34 %	61.1 % $\pm$ 3.2	33.4 % $\pm$ 1.6
14 %	90.7 % $\pm$ 0.4	67.9 % $\pm$ 3.0

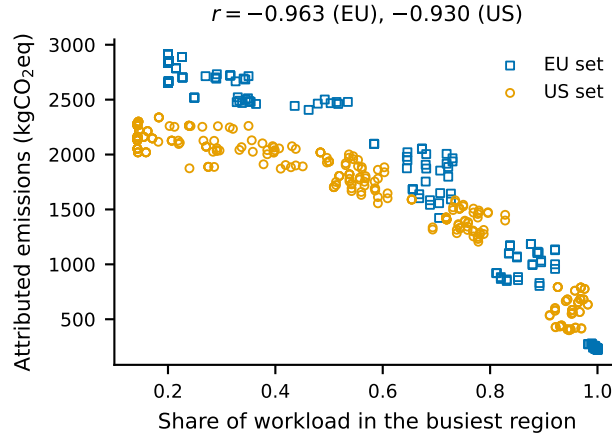


Figure 6: The mechanism common to all three findings. Each point is one experimental cell. Emissions fall as a larger share of the workload is placed in the busiest region, and the relationship holds across both region sets and every capacity, deadline margin and policy examined.

Power model. Repeating the entire sweep under the linear 120/480 W model in place of the measured curve raises absolute emissions by approximately 40 % but leaves every reported proportion within its sampling spread: the capacity effect moves from 77.4 % $\pm$ 2.3 to 75.9 % $\pm$ 2.2, the margin effect at 82 % utilisation from 0.9 % $\pm$ 0.6 to 1.0 % $\pm$ 0.5, the correlation with origin intensity from [+0.941, +0.998] to [+0.943, +0.998], and the EU-to-US ratio at the lowest utilisation from 90.7/67.9 to 90.1/67.1. The conclusions of Sections 5.1–5.3 are therefore not artefacts of the power model adopted.

The absolute values are not invariant, and the direction of the change is worth recording. Under the measured curve, round-robin placement rises by 44 % while the carbon-aware policies rise by 26 % or less, so the reduction attributed to space shifting increases rather than decreases. The attributed basis charges the full draw of every host carrying load, and the measured machine idles at 55 % of peak rather than 25 %; raising the fixed cost of activating a host penalises the policy that activates the most. Round-robin placement leaves 72 hosts active at the lowest utilisation against 47 for space shifting and 38 for space+time. Part of what the attributed basis records as a carbon-aware saving is therefore a consolidation saving. The dynamic basis, which excludes idle draw entirely, separates the two: on that basis the same comparison yields reductions of 64 % and 71 % respectively, confirming that regional placement rather than consolidation accounts for the larger share.

6 Discussion

6.1 What these results imply for reporting

The three parameters examined here are not exotic. Each is fixed in the course of designing any evaluation of carbon-aware scheduling, and each is ordinarily left unstated. Their combined effect, in our experiments, spans reported reductions from 4.6 % to 90.7 % for the same class of policy. A figure reported without them is therefore not comparable with a figure reported elsewhere, and the divergence in the literature is to that extent expected rather than anomalous.

Four items would make such figures comparable.

Capacity utilisation. Total demand as a fraction of total available slot-hours, stated alongside any claim about deadline flexibility. Without it, a reduction attributed to flexibility cannot be distinguished from a reduction attributable to an unconstrained pool.

Origin region. For any policy that shifts in time but not in space, the region in which the workload originates, and preferably the distribution of results across candidate origins rather than a single value.

Region-set composition. The candidate regions and their mean carbon intensities, from which a reader can judge whether the set admits a dominant optimum. The ratio between the cleanest region and the second cleanest is a compact summary.

Accounting basis. Whether reported emissions include the idle draw of provisioned hosts, of active hosts only, or neither. As Section 3.4 shows, the first of these cannot distinguish policies at all in a fixed-fleet setting, and the second conflates regional placement with consolidation.

6.2 Concentration and deployability

The mechanism common to all three findings is concentration. Across every valid cell in both region sets, the correlation between the share of work placed in the busiest region and attributed emissions is below -0.93 . The policies reduce emissions by moving work into the cleanest reachable grid, and the three parameters govern how much of the workload can be moved, where a constrained policy starts from, and how much is gained by arriving.

This has an uncomfortable consequence. The configurations that report the largest reductions are those in which 95 % or more of the workload is placed in one region, and that is not a configuration any operator could adopt. Real deployments are constrained by data residency, by latency to users, by capacity that is finite in exactly the way our capped configurations model, and by the grid-level effects of concentrating demand. The reductions reported at low utilisation should therefore be read as upper bounds obtained under a placement distribution that would not survive contact with those constraints, rather than as achievable savings.

Read the other way, the result is more encouraging than it first appears. The 82 % utilisation row, where deadline flexibility is worth 0.9 % and space shifting 4.6 %, is the row that most resembles an efficiently operated facility. That the returns are small there is a finding about where effort is best directed, not an argument against carbon-aware scheduling as such.

6.3 On the interpretation of time shifting

Section 5.2 shows that a time-shifting policy's outcome tracks the mean intensity of its origin rather than that origin's temporal structure. The natural reading is that temporal flexibility is worth little. The more precise reading is that temporal flexibility is worth little *when spatial flexibility is available and capacity is not binding*, which is the regime in which the two are usually compared. Our delay-insensitive robustness condition supplies the counterexample: at tight capacity and wide margin, where relocation is constrained and deferral is not, the correlation with origin intensity falls to $+0.403$ and that with diurnal amplitude rises to $+0.776$. Temporal structure becomes the dominant term exactly when it is the only term available.

A second observation from that condition bears on the premise of deadline-based scheduling. The workload class the provider designates as delay-insensitive has a mean lifetime an order of magnitude longer than the general mixture, so that a given absolute margin represents a far smaller fraction of runtime. Deadline margins expressed in hours are not comparable across workload classes, and margins expressed as a fraction of runtime would be a more meaningful axis for future studies.

7 Limitations

Perfect foresight. Planning is performed against the ground-truth carbon trace. This isolates the parameters under study from prediction error, but it means the absolute reductions reported here are upper bounds with respect to forecast quality. Whether the three effects survive realistic forecast error is a question we address separately.

Single workload source. All virtual machine requests derive from one provider’s 2019 trace. The duration and size distributions of that trace shape the capacity utilisations at which our effects appear, and a workload with a different shape would place the transition points elsewhere, though we have no reason to expect the direction of the effects to change.

Deadline margins are imposed rather than observed. The trace records a workload category but not a service-level objective, so deadline margins are assigned as an experimental factor rather than read from the data. The delay-insensitive condition is the closest available approximation to a provider-designated flexibility class.

No migration or preemption. Virtual machines run to completion where they start, following the formulation of the work we build on. Policies permitting mid-flight migration would have a larger action space, and the capacity interaction we report might take a different form under them.

Excluded costs. Network transfer between regions, the embodied carbon of the hardware, and datacentre overhead beyond the host itself are not modelled. The first of these is of particular relevance given that the largest reductions we observe coincide with the greatest concentration of load.

Two region sets. The claim of Section 5.3 rests on a contrast between two sets. The direction of the effect follows from the mechanism and we expect it to generalise, but the coefficient relating region-set composition to reported savings is not established by two points.

8 Conclusion

The reductions reported for carbon-aware virtual machine scheduling depend on three parameters that evaluations do not customarily state. Capacity utilisation determines whether deadline flexibility has any value at all, varying its worth from 32 % to under 1 % across the range we examine. The origin region assigned to a time-shifting policy determines that policy’s result more completely than the temporal structure it is designed to exploit, with a spread across candidate origins of a factor of 3.6. The composition of the candidate region set determines how much a spatial policy can gain, with one set reporting twice the reduction of another under identical conditions. All three operate through the same mechanism, the concentration of work into the cleanest reachable grid, and the configurations that report the largest reductions are those in which nearly the entire workload is placed in a single region. We recommend that capacity utilisation, origin region, region-set composition and carbon accounting basis be reported alongside any figure for carbon-aware scheduling, and we note that the returns are smallest in precisely the high-utilisation regime that an efficiently operated facility occupies.

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